

Document Name : AiSWEI API Manual				
Doc. No.	1.0	Originating Dept.	Application and network department	
Rev. No.	1.0	Total Pages	19	

Revision History

Rev. No.	Change Description	Author	Reviewed By	Date
V1.0	First version	Guodong.shen		2020-08-20

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1. Brief Introduction

1.1 Purpose

AiSWEI API supply access to get data of PV plant, you can get data from AiSWEICloud by this API, and embed them in your application or website, to customize your personalized presentation.

This document will explain how to use the APIs, for developer and maintainer.

1.2 Abbreviations and Definitions

AiSWEICloud --Visual monitoring system for customer's PV plant supplied by AiSWEI

API -- Application Programmer's Interface

JSON -- JavaScript Object Notation

HTTP -- Hypertext Transfer Protocol

RFC -- Request for Comment

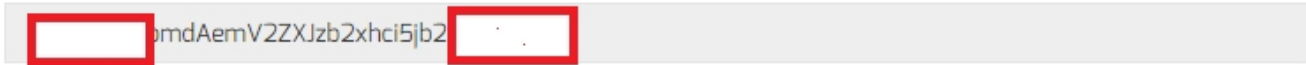
RPC -- Remote Procedure Call

APP key, APP secret

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Key -- plant api key。 It is used to identify the request id of the plant. You can find in on the web page of the AiSWEICloud. (Navigation bar:"Configuration" -> "Plant Configuration" ->"5. API key")

5 • Api Key



1.3 Disclaimer

AiSWEI API is supplied by AiSWEICloud, which is a free value-added service. You have to bear your own risk and loss associated with using this service.

There may be different between the content this document describe and actual use, because of version updating. We may update the version without prior notice, please pay your attention to our website. In use process, you can contact with us if any question.

1.4 Reference

- RFC 4627 :The application/json Media Type for JavaScript Object Notation (JSON)
- JSON-RPC 1.1 Working Draft August 2006<http://json-rpc.org/wd/JSON-RPC-1-1-WD-20060807.html>
- Introducing JSON<http://www.json.org/>

1.5 Limits

- API traffic limit : 100 times per minute

2. API Description

2.1 Request

2.1.1 Request Path

<http://e710888d3ccb4638a723ff8d03837095-cn-qingdao.aliapi.com/demo/post>

2.1.2 Request Method

POST

2.1.3 Request Body

FormParam1=FormParamValue1&FormParam2=FormParamValue2

2.1.4 Request Head

- **Host:** e710888d3ccb4638a723ff8d03837095-cn-qingdao.aliapi.com
- **Date:** Mon, 22 Aug 2016 11:21:04 GMT
- **User-Agent :** Apache-HttpClient/4.1.2 (java 1.6)
- **Content-Type :** application/x-www-form-urlencoded; charset=UTF-8

// request body type, please set according to the actual request body content.

- **Accept:** application/json

// Request response type, part of the API can be based on the specified response type to return to the corresponding data format, it is recommended to manually specify the request header, if not set, part of the HTTP client will set the default value * / *, resulting in a signature error.

➤ **X-Ca-Request-Mode:** debug

//Whether to open the Debug mode, case insensitive, do not set the default off, the general API debugging phase can open this setting.

➤ **X-Ca-Version:** 1

// API version number, all APIs currently only support version number "1", you can not set this request header, the default version number is "1".

➤ **X-Ca-Signature-Headers:** X-Ca-Request-Mode,X-Ca-Version,X-Ca-Stage,X-Ca-Key,X-Ca-Timestamp

//The client will participate in the signature of the custom request header, the server will be based on this configuration to read the request header to sign, where the settings do not contain Content-Type, Accept, Content-MD5, Date request header, these request head has been included in the basic signature In the structure, refer to the request signature documentation for details.

➤ **X-Ca-Stage:** RELEASE

// Request the API of the Stage, currently supports TEST, PRE, RELEASE three Stage, case insensitive, API providers can choose to release to which stage, only published to the designated stage after the API can be called, otherwise it will prompt the API can not find Or Invalid Url.

➤ **X-Ca-Key:** 60022326

// Request the AppKey, please go to the API Gateway console generated, only access to API authorization can be called after the cloud through the market and other channels to buy the API has been granted acquiescence APP, Ali cloud all cloud products share a set of AppKey system, remove AppKey Please be careful to avoid affecting other cloud products that have already been opened.

➤ **X-Ca-Timestamp:** 1471864864235

// The timestamp of the request is the number of milliseconds of the current time, that is, from January 1, 1970 to the present time, the time stamp is valid for 15 minutes.

➤ **X-Ca-Nonce:**b931bc77-645a-4299-b24b-f3669be577ac

// Request unique identifier, within 15 minutes AppKey + API + Nonce can not be repeated, combined with the timestamp to play a role in anti-replay.

➤ **X-Ca-Signature:** FJleSrCYPGCU7dMLTG+UD3Bc5Elh3TV3CWHtSKh1Ys=

// Request signature.

➤ **CustomHeader:** CustomHeaderValue

// Custom request header, here only as an example, the actual request can be set according to the API definition of multiple custom request header.

2.2 Response

2.2.1 Status Code

➤ 400

// Response status code, greater than or equal to 200 less than 300 means successful; greater than or equal to 400 less than 500 for the client error; greater than 500 for the server error.

2.2.2 Response Head

➤ **X-Ca-Request-Id:** 7AD052CB-EE8B-4DFD-BBAF-EFB340E0A5AF

// Request the unique ID, the request once the API gateway application, the API gateway will generate the request ID and return to the client through the response header, the proposed client and back-end services are recorded this request ID, can be used for troubleshooting and tracking.

➤ **X-Ca-Error-Message:** Invalid Url

// The API gateway returns an error message that the API gateway returns the error message to the client through the response header when an error occurs.

➤ **X-Ca-Debug-Info:** {"ServiceLatency":0,"TotalLatency":2}

// When you open the Debug mode will return to the Debug information, this information may be changed later, only for the FBI phase reference

When you call the API, you need to include the signature information in the request, whether you are using HTTP or the HTTPS protocol. AppKey is used to identify you, AppSecret is the key used to encrypt the signature string and the server-side validation signature string. For details on how to encrypt a signed signature, see the document [4.request signature](#).

3. API List

N O	Groupin g	Key pair	API name	Descri ption
1	OLD API	Please contact with the salesman or service.	getPlantOverview	To get overview information data of plant
2			getPlantOutput	To get power or energy data of plant
3			getPlantEvent	To get error event of plant
4			getInverterOverview	To get all inverter overview information data of plant

			getplanlist	To get all plants information under the user's account.
			getdevicelist	To get all devices information
5	ODMAP I	Please contact with the salesman or service.	bindpmu	To bind the monitor to the plant
6			getpmudata	To get the inverter data which controlled by monitor
7			unbindpmu	To unbind the monitor from the plant

➤ Traffic Control:

- 1) OLDAPI: 100 times/Minutes
- 2) DOMAPI: 50 times/Minutes

3.1 [getPlantOverview](#)

- Request Path : [https:// api.general.aisweicloud.com/getPlantOverview](https://api.general.aisweicloud.com/getPlantOverview)
- HTTP Method: GET
- Request Parameters:

Name	Type	Description
key	String	The key of your plant

- Response Type:JSON
- Response Parameters:

Name	Type	Description
sid	String	The ID of requested plant
E-Today	String	Energy generated today
E-Month	String	Energy generated this month
E-Total	String	Energy generated of total
TotalYield	String	Total yield
CO2Avoided	String	Carbon dioxide emissions

ludt	String	The last update time
Power	String	Current power
status	String	0:offline; 1:normal; 2:warning(at least a device offline); 3:error(inverter errors happened in one hour)

➤ Response Example

```
{
  "sid":812,"
  ludt":"2018-01-29 18:46:56",
  "status":"0",
  "E-Today":{"unit":"KWh","value":0},
  "E-Month":{"unit":"KWh","value":0},
  "E-Total":{"unit":"MWh","value":186.45},
  "TotalYield":{"unit":"€","value":68985.59},
  "CO2Avoided":{"unit":"T","value":149.16},
  "Power":{"unit":"W","value":0}
}
```

3.2 getPlantOutput

➤ Request Path : [https:// api.general.aisweicloud.com/getPlantOutput](https://api.general.aisweicloud.com/getPlantOutput)

➤ HTTP Method: GET

➤ Request Parameters:

Name	Type	Description
key	String	The key of your plant
period	String	Options: bydays : To get power data of one day bymonth : To get energy generated data of one month byyear : To get energy generated data of one year bytotal: To get energy generated data of all years
date	String	Depend on “period”you should give a date format which correspond to “period”. Date : yyyy-MM-dd Month : yyyy-MM Year : yyyy It's no need to give this parameter if “period” is “bytotal”.

➤ Response Type:JSON

➤ Response Parameters:

Name	Type	Description
dataunit		Unit

sid	String	The ID of requested plant
data	String	Every "data" composed of the following: Time: No : Serial number order by time Value: power

➤ Response Example:

● bydays

```
{
  "sid":893,
  "dataunit":"KW",
  "data":
  [
    {"time":"00:00","no":"0","value":"0.0"},
    {"time":"00:20","no":"2","value":"0.0"},
    {"time":"00:40","no":"4","value":"0.0"},
    {"time":"01:00","no":"6","value":"0.0"},
    {"time":"01:20","no":"8","value":"0.0"},
    {"time":"01:40","no":"10","value":"0.0"},
    .....
    {"time":"23:40","no":"142","value":"0.0"}
  ]
}
```

● bymonth

```
{
  "dataunit":"KWh",
  "sid":"36",
  "data":
  [
    {"time":"2014-04-01","no":"1","value":"0.0"},
    {"time":"2014-04-02","no":"2","value":"0.0"},
    {"time":"2014-04-03","no":"3","value":"0.0"},
    ...
    {"time":"2014-04-28","no":"28","value":"0.0"},
    {"time":"2014-04-29","no":"29","value":"0.0"},
    {"time":"2014-04-30","no":"30","value":"0.0"}
  ]
}
```

● byyear

```
{
  "dataunit":"KWh",
  "sid":"36",
  "data":
  [
    {"time":"2014-01","no":"1","value":"0.0"},
    {"time":"2014-02","no":"2","value":"0.0"},
    {"time":"2014-03","no":"3","value":"0.0"},
    {"time":"2014-04","no":"4","value":"0.0"},
    ...
  ]
}
```

```

        {"time":"2014-11","no":"29","value":"0.0"},
        {"time":"2014-12","no":"30","value":"0.0"}
    ]
}

```

- bytotal


```

      {
        "dataunit":"MWh",
        "sid":"36",
        "data":
        [
          {"time":"2012","no":"1","value":"4.069"},
          {"time":"2013","no":"2","value":"0.308"}
        ]
      }

```

3.3 getPlantEvent

➤ Request Path : <https://api.general.aisweicloud.com/getPlantEvent>

➤ HTTP Method: GET

➤ Request Parameters:

Name	Type	Description
key	String	The key of your plant
sdt	String	The start date
edt	String	The end date (Should not more than 7 days)

➤ Response Type:JSON

➤ Response Parameters:

Name	Type	Description
sid	String	The ID of requested plant
ssno	String	Serial number of inverter
eventCode	String	Event code
eventType	String	Event type 3 : error type
eventTime	String	The time when event occur

➤ Response Example

```

{
  "sid":"36",
  "data":

```

```
[
  {"ssno":"8882000A10902151","eventCode":"135","eventType":"3","eventTime":"2013-06-17
06:58:14.0"}
  {"ssno":"8882000A10902151","eventCode":"135","eventType":"3","eventTime":"2013-06-17
07:04:14.0"}
]
```

3.4 [getInverterOverview](#)

- Request Path: <https://api.general.aisweicloud.com/getInverterOverview>
- HTTP Method: GET
- Request Parameters:

Name	Type	Description
key	String	The key of your plant

- Response Type:JSON
- Response Parameters:

Name	Type	Description
isno	String	Serial number of inverter
e_today	String	The inverter generating capacity today
e_month	String	The inverter generating capacity in this month
e_total	String	The inverter generating capacity in this year
co2	String	The inverter co2 discharge
yield	String	The inverter total revenue

- Response Example

```
{
  "sid": 6123,
  "data":
  [
    {"isno": "ZP033K0011640095","e_today": "3.2","e_month": "1675.3","e_total": "20054.9","co2":
    "16234.44","yield": "16356.77"},
    {"isno": "ZP033K0011640061","e_today": "1.7","e_month": "2844.59","e_total":
    "16724.79","co2": "13538.72","yield": "13640.74"},
    {"isno": "ZP033K0011630006","e_today": "19.39","e_month": "1903.69","e_total":
    "19483.79","co2": "15772.13","yield": "15890.98"},
    {"isno": "ZP033K0011630037","e_today": "12.5","e_month": "1397.3","e_total": "15491.2","co2":
    "12540.12","yield": "12634.62"}
  ]
}
```

}

3.5 getplantlist

- Request Path: <https://api.general.aisweicloud.com/planlist>
- HTTP Method: GET
- Request Parameters

Name	Type	Description
token	String	The token of user's account. Please contact with the salesman or service of SMA(China).
page	String	Page number. We will page the plants with 20 plants/page in default, if there are lots of plants. Page number starts from one.
size	String	Maximun plant amount. 20 plants/ page in default.

- Response Type: JSON (application/json; charset=utf-8)
- Response Parameters

Name	Type	Description
totalcount	Integer	Plants amount
pagesize	Integer	Maximun plant amount. 20 plants/ page in default.
page	Integer	Page number
country	Integer	Country code of the plant
totalpower	Integer	Installed capacity
apikey	Integer	Plant api key
city	String	City name of the plant
ludt	String	Last data uploading time of the plant
etoday	Double	Energy from inverters on current day.
created	String	The Creation time of the plant
wd	Double	Latitude of the plant
etotal	Double	Total energy of the plant from creation.
imgurl	String	The url of the image

province	Integer	Province code
name	String	Plant name
jd	Integer	Longitude of the plant
status	Integer	Plant status:0:all devices offline;1:all devices online and no error ;2:At least one device offline;3:At least one device malfunctions °

➤ Response Example

```
{
  "data": {
    "totalcount": 23,
    "pagesize": 20,
    "page": 1,
    "list": [
      {
        "country": "XXXX",
        "totalpower": "XXXXX",
        "apikey": "XXXXXXXXXXXXXXXXXXXXXXXXXXXX",
        "city": "XXXXXXX",
        "ludt": "2018-12-06 11:44:04",
        "etoday": "XXXX.X",
        "createdt": "2012-08-31 21:10:45",
        "wd": "XX.XXXXXX",
        "etotal": "XXXXX.XX",
        "imgurl": "https://www.aisweicloud.com/upload/station/defaultBg.jpg",
        "province": "XXXX",
        "name": "XXXXXXXXXX",
        "jd": "XX.XXXXXX",
        "status": 2
      },
      .....
    ]
  }
}
```

3.6 getdevicelist

- Request Path: <https://api.general.aisweicloud.com/devicelist>
- HTTP Method: GET
- Request Parameters

Name	Type	Description
------	------	-------------

key	String	The key of your plant
-----	--------	-----------------------

➤ Response Type: JSON (application/json;charset=utf-8)

➤ Response Parameters

Name	Type	Description
pstate	Integer	Status of the Monitor.0:offline; 1:online
psn	String	Serial number of the monitor
isn	String	Serial number of the inverter
ludt	String	Last updating time of the inverter
istate	Integer	Status of the inverter.0:offline;1:online;2:Data updating;3:special status as offline.

➤ Response Example

```
{
  "data": {
    "list": [
      {
        "pstate": 1,
        "inverters": [
          {
            "isn": "xxxxxxxx",
            "ludt": "2018-12-07 10:40:25",
            "istate": 1
          }
        ],
        "psn": "xxxxxxxx"
      },
      .....
    ]
  }
}
```

3.7 bindpmu

➤ Request Path: http://api.odm.zevercloud.cn/bindpmu

➤ HTTP Method: GET

➤ Request Parameters:

Name	Type	Description
------	------	-------------

apikey	String	The key of your plant
psno	String	Serial number of monitor
idex	String	Register key of monitor

➤ Response Type:JSON

➤ Response Parameters:

Name	Type	Description
info	String	Result message

➤ Response Example

```
{"info":"bind success"}
```

3.8 unbindpmu

➤ Request Path: [http:// api.odm.zevercloud.cn / unbindpmu](http://api.odm.zevercloud.cn/unbindpmu)

➤ HTTP Method: GET

➤ Request Parameters:

Name	Type	Description
apikey	String	The key of your plant
psno	String	Serial number of monitor
idex	String	Register key of monitor

➤ Response Type:JSON

➤ Response Parameters:

Name	Type	Description
info	String	Result message

➤ Response Example

```
{"info":"unbind success"}
```

4. Request a signature

4.1 Domain Name

- Each API belongs to an API group, and each API group has a unique domain name. These independent domain names are bound by the service provider. API Gateway uses a domain name to locate an API group.
- The domain name is in the format of **www.[Independent domain name].com/[Path]?[HTTPMethod]**. At the public beta stage, the API user needs to obtain this domain name offline from the API service provider.
- Alibaba Cloud API Gateway uses the domain name to locate a unique API group, and then locate the unique API through Path+HTTPMethod.
- You must obtain API documentation in the deprecation environment from the API service provider. This documentation must include necessary parameter information, such as the domain name and path.

4.2 System level header

- [Required] X-Ca-Key: AppKey
- [Required] X-Ca-Signature: Signature string
- [Optional] X-Ca-Timestamp: The time stamp in milliseconds passed by the API caller, that is, the milliseconds of the time from January 1, 1970 until now. By default, it is valid within 15 minutes.
- [Optional] X-Ca-Nonce: The UUID generated by the API caller. This header is used with the time stamp to prevent replay.
- [Optional] Content-MD5: When the request body is not a Form, calculate the MD5 value of the body and send that value to the cloud gateway for checking.
- [Optional] X-Ca-Stage: The stage where the requested API belongs. Only test and release are supported, and the default value is release.

4.3 Signature verification

4.3.1 Prepare APPKey

Appkey, call the API identity, you can contact the service provider to apply

4.3.2 The organization participates in the signature calculation of the string

```
1. String stringToSign=  
2. HTTPMethod + "\n" +  
3. Accept + "\n" +  
4. Content-MD5 + "\n"  
5. Content-Type + "\n" +  
6. Date + "\n" +  
7. Headers +  
8. Url
```

Each letter of the HTTPMethod value must be capitalized.

If Accept, Content-MD5, Content-Type, and Date are empty, add a linefeed (\n). If Headers is empty, a linefeed (\n) is not required. The specified Headers includes a linefeed (\n). For more information, see the headers organization method described as follows.

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Content-MD5:

Content-MD5 indicates the MD5 value of the body. MD5 is only calculated when the body is not a Form. The calculation method is as follows:

```
1. String content-MD5 = Base64.encodeBase64(MD5(bodyStream.getBytes("UTF-8")));
```

The **bodyStream** indicates the byte array.

Headers:

Headers indicates the keys and values of the headers involved in signature calculation. Note that X-Ca-Signature and X-Ca-Signature-Headers are excluded in Headers signature calculation.

Headers organization method:

Rank the keys of all Headers involved in signature calculation in lexicographic order and then splice them in the following method:

```
1. String headers =  
2. HeaderKey1 + ":" + HeaderValue1 + "\n"+  
3. HeaderKey2 + ":" + HeaderValue2 + "\n"+  
4. ...  
5. HeaderKeyN + ":" + HeaderValueN + "\n"
```

Url

URL indicates the Form parameter in the Path+Query+Body. The organization method is as follows:

Rank the keys of Query+Form in lexicographic order and then splice them in the following method. If Query or Form is empty, the URL is equal to Path, and a question mark (?) is not required to be added.

```
1. String url =  
2. Path +  
3. "?" +  
4. Key1 + "=" + Value1 +  
5. "&" + Key2 + "=" + Value2 +  
6. ...  
7. "&" + KeyN + "=" + ValueN
```

Note that Query or Form may have multiple values. If multiple values exist, use the first value for signature calculation.

4.3.3 Calculate the signature

```
1. Mac hmacSha256 = Mac.getInstance("HmacSHA256");  
2. byte[] keyBytes = secret.getBytes("UTF-8");  
3. hmacSha256.init(new SecretKeySpec(keyBytes, 0, keyBytes.length, "HmacSHA256"));  
4. String sign = new String(Base64.encodeBase64(hmacSha256.doFinal(stringToSign.getBytes("UTF-8")), "UTF-8"));
```

The **secret** indicates the key corresponding to an app..

4.3.4 Pass the signature

Put the calculated signature in the Header of the Request. The key is X-Ca-Signature.

Separate the keys of all Headers involved in signature calculation by commas and put them in the Header of the Request regardless of the order. The key is X-Ca-Signature-Headers.

For more information about the demo of signature calculation, click [here](#).

4.3.5 Signature error detection method

When the signature verification fails, the API gateway will be the server's StringToSign into the HTTP Response Header to return to the client, Key: X-Ca-Error-Message, only the local calculation of the StringToSign and the server back to the StringToSign To be able to find a problem;

If the server is consistent with the client's StringToSign, check that the key used for the signature calculation is correct;

Because the HTTP Header can not represent a newline, the string break in StringToSign is filtered out, and the line breaks are ignored when compared.

4.3.6 Signature demo

Signature demo detailed demo (JAVA).

Please refer to the link: <https://github.com/aliyun/api-gateway-demo-sign-java>.

5. Event Code Description table

Error Code	Error Description
101	SCI Fault
102	EEPROM R/W Fault
103	RLY-Check Fault
104	DC INJ. High
105	AUTO TEST FAILED
106	High DC Bus
107	Ref.Voltage Fault
108	AC HCT Fault
109	GFCI Fault
110	Device Fault
111	M-S version unmatched
112	Reserve
132	Reserve
133	Fac Fault
134	Vac Fault
135	Utility Loss
136	Ground fault
137	PV Over voltage
138	ISO Fault
139	Reserve
140	Over Temp.
141	Vac differs for M-S
142	Fac differs for M-S

143	Groud I differs for M-S
144	DC inj. differs for M-S
145	Fac,Vac differs for M-S
146	High DC Bus
147	Consistent Fault
148	Average volt of 10 minutes Fault
149	Reserve
150	Reserve
151	
152	Fuse Fault
153	ISO check: before enable constant current , ISO voltage> 300mV
154	ISO check: after enable constant current , ISO voltage out of range (1.37v +/- 20%)
155	ISO check: N P relay change , ISO voltage sudden below 40mV
156	GFCI protect fault :30mA lever
157	GFCI protect fault :60mA lever
158	GFCI protect fault :150mA lever
159	PV1 string current abnormal
160	PV2 string current abnormal
161	DRED Communication Fails (S9 open)
162	Operate the disconnection device (S0 close)